

RAG Chunking Test Corpus

A benchmark document for evaluating retrieval-augmented generation
chunk strategies on voice-agent knowledge bases

Test corpus · Version 1.0
Prepared for the Tone platform ingestion pipeline

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1. Introduction to conversational voice agents

A conversational voice agent is a software system that holds spoken, real-time interactions with a caller. It sits behind a phone number or a browser WebRTC endpoint and, from the moment the far end says hello, orchestrates a tightly-coupled pipeline: an ingress transport that delivers audio frames, a speech-to-text (STT) engine that converts those frames to tokens, a large language model (LLM) that decides what to say next, a text-to-speech (TTS) engine that renders the reply back into audio, and finally an egress transport that ships the audio back to the caller with the lowest achievable latency. When each stage is measured in tens of milliseconds, the aggregate feels like a conversation. When any single stage stalls, the caller notices and the illusion breaks.

Building this pipeline used to be the exclusive domain of large contact-centre vendors whose product suites started at six figures per year and whose onboarding involved months of professional-services engagements. The rise of hosted STT and TTS APIs, together with commodity LLM inference, has collapsed that barrier. A modern voice-agent platform lets a product manager assemble a working agent in an afternoon: pick a voice, pick a language model, write a prompt, wire it to a phone number, and ship. The interesting engineering problems have moved from the DSP layer up into orchestration, evaluation, and knowledge retrieval.

Retrieval-augmented generation, or RAG, is the term of art for the pattern where the LLM consults a knowledge base at inference time before composing a reply. In a voice-agent context the knowledge base typically contains product documentation, internal SOPs, help-desk articles, and policy documents. When a caller asks "how do I cancel my plan," the agent does not answer from the model's training data; it embeds the query, retrieves the most similar chunks from the knowledge base, and asks the model to answer grounded in those chunks. Latency budgets are brutal — a full retrieval round-trip must complete inside roughly two hundred milliseconds — so every architectural choice, from the parser to the chunker to the embedding provider to the vector store, has to earn its place.

This document exists to give the ingestion pipeline something interesting to chew on. It mixes long-form prose, tabular data, dialogue-shaped transcripts, code, and short-form reference material on purpose. Each chunker strategy in the pipeline will produce a subtly different partition; comparing them side by side is the fastest way to build intuition about which strategy suits which corpus.

2. Reference pipeline architecture

The reference pipeline used throughout this corpus is intentionally boring, because voice agents fail interestingly enough on their own. Audio ingress terminates at either a Daily WebRTC room or a Twilio Media Streams websocket. Frames are decoded, resampled to sixteen kilohertz, and

pushed into the STT engine as they arrive. The STT engine streams interim transcripts back; a voice-activity detector marks turn boundaries; and once a turn closes the finalised transcript is handed to the LLM together with any retrieved context. The LLM streams tokens back into the TTS engine, which in turn streams audio frames back to the transport without waiting for the full utterance to complete. This end-to-end streaming is what gives modern voice agents their eerie responsiveness.

Component summary

Stage	Reference component	Alternatives	Latency budget (p50)
Ingress transport	Daily WebRTC	Twilio, LiveKit, raw WS	20 ms
Speech-to-text	Deepgram Nova-2	Whisper (self-hosted), Azure Speech, AssemblyAI	250 ms
Language model	GPT-4o mini	Claude 3.5 Haiku, Llama 3.1 8B (vLLM)	400 ms
Retrieval	pgvector + text-embedding-3-small	Pinecone, Qdrant, Milvus	120 ms
Text-to-speech	ElevenLabs Turbo v2	Cartesia Sonic, PlayHT, Azure Neural	280 ms
Egress transport	same as ingress	—	20 ms

3. Provider comparison

The following comparison is drawn from internal benchmarks against a fixed one-thousand-call regression suite. Latencies are median values measured from the caller's perspective. Cost is a representative per-minute figure at a monthly commit tier; discount tiers move the numbers substantially. Accuracy is expressed as word error rate for STT and as an internal five-point listener rating for TTS. Numbers should be read as directional, not gospel.

Category	Provider	Latency (p50)	Latency (p95)	Cost / min	Accuracy / MOS
STT	Deepgram Nova-2	210 ms	430 ms	\$0.0043	WER 6.1%

Category	Provider	Latency (p50)	Latency (p95)	Cost / min	Accuracy / MOS
	Whisper large-v3 (self-hosted A10G)	340 ms	820 ms	\$0.0025	WER 5.4%
	Azure Speech en-US neural	260 ms	510 ms	\$0.0067	WER 6.9%
LLM	GPT-4o mini	380 ms	720 ms	\$0.0090	—
	Claude 3.5 Haiku	410 ms	780 ms	\$0.0095	—
	Llama 3.1 8B Instruct (vLLM, L40S)	290 ms	640 ms	\$0.0034	—
TTS	ElevenLabs Turbo v2	240 ms	460 ms	\$0.0180	MOS 4.4
	Cartesia Sonic	180 ms	370 ms	\$0.0165	MOS 4.2
	PlayHT Play-3	320 ms	590 ms	\$0.0140	MOS 4.1

4. Deployment cost worksheet

The worksheet below assumes a small-business tier: five thousand calls per month with a mean call duration of ninety seconds. Costs are annualised for readability; divide by twelve for the monthly view. The "self-hosted" row assumes an existing Kubernetes footprint with GPU capacity — capex is not amortised into the figures.

Line item	Unit cost	Volume / month	Monthly cost	Annual cost
Ingress + egress transport	\$0.004 / min	7,500 min	\$30	\$360
STT (Deepgram Nova-2)	\$0.0043 / min	7,500 min	\$32.25	\$387
LLM (GPT-4o mini, avg 800 tok/turn)	\$0.15 / 1M tok	4M tok	\$60	\$720
TTS (ElevenLabs Turbo v2)	\$0.018 / min	7,500 min	\$135	\$1,620
Vector DB (pgvector, RDS db.m6g.large)	flat	—	\$121	\$1,452

Line item	Unit cost	Volume / month	Monthly cost	Annual cost
Compute (2 × EKS m6i.xlarge)	flat	—	\$281	\$3,372
Total	—	—	\$659.25	\$7,911

Rule of thumb: TTS and LLM together consume roughly 30% of your run rate at low volume and cross 55% at high volume. Renegotiating those two contracts is almost always higher leverage than infrastructure re-platforming.

5. Sample call transcripts

Two representative transcripts follow. They are intentionally short and structured; the sentence chunker should keep individual speaker turns intact, while the semantic chunker should merge topically related exchanges.

5.1 Order status inquiry

Agent: Thanks for calling Northwind Supply. How can I help you today?

Caller: Hi, I placed an order on Tuesday and I still don't see a shipping confirmation. Can you check on it?

Agent: Absolutely. Do you have the order number handy, or should I look it up by the email on file?

Caller: The order number is NW-7742-A.

Agent: One moment. Alright, I see order NW-7742-A for two pallets of the industrial degreaser. It went into fulfilment yesterday morning and is scheduled to ship this afternoon from our Ohio warehouse. You should see a UPS tracking number in your inbox by six PM Eastern.

Caller: That's great, thanks. Do you know when it'll arrive?

Agent: Since you're in Zone 4 the standard transit time is three business days, so you're looking at delivery on Monday. If you need it sooner I can bump it to two-day air for an additional forty-two dollars.

Caller: Monday's fine. Thank you.

5.2 Plan cancellation

Agent: Hi, this is Vega speaking. Who am I connecting with?

Caller: Marcus Delaney. I want to cancel my subscription.

Agent: I'm sorry to hear that, Marcus. Before I process the cancellation, can I ask what prompted the decision?

Caller: Honestly, I just don't use it enough to justify the cost.

Agent: Understood. We do have a lower-tier plan at nine dollars a month that keeps your account and history active. Would that be a better fit, or would you prefer to go ahead with the full cancellation?

Caller: Let's do the full cancellation, please.

Agent: Done. Your account will remain active until the end of the current billing period, October 27, and you'll retain read-only access to your history for ninety days after that. You'll receive a confirmation email within the next few minutes. Anything else I can help with?

Caller: No, that's it. Thanks.

6. Configuration reference

The examples below show a typical agent configuration in two formats. Code blocks are deliberately long enough to test whether the chunker preserves indentation and quoting.

6.1 JSON

```
{
  "agent_id": "agt_9a3f",
  "display_name": "Northwind Support Bot",
  "language": "en-US",
  "stt": {
    "provider": "deepgram",
    "model": "nova-2",
    "params": { "smart_format": true, "punctuate": true, "interim_results": true }
  },
  "llm": {
    "provider": "openai",
    "model": "gpt-4o-mini",
    "params": { "temperature": 0.4, "max_tokens": 400, "top_p": 0.95 }
  },
  "tts": {
    "provider": "elevenlabs",
    "voice_id": "Rachel",
    "params": { "model_id": "eleven_turbo_v2", "stability": 0.55 }
  },
  "retrieval": {
    "knowledge_base_id": "kb_5511",
    "top_k": 6,
    "score_threshold": 0.72
  }
}
```

```
}  
}
```

6.2 YAML

```
agent_id: agt_9a3f  
display_name: Northwind Support Bot  
language: en-US  
stt:  
  provider: deepgram  
  model: nova-2  
  params:  
    smart_format: true  
    punctuate: true  
    interim_results: true  
llm:  
  provider: openai  
  model: gpt-4o-mini  
  params:  
    temperature: 0.4  
    max_tokens: 400  
    top_p: 0.95  
tts:  
  provider: elevenlabs  
  voice_id: Rachel  
  params:  
    model_id: eleven_turbo_v2  
    stability: 0.55  
retrieval:  
  knowledge_base_id: kb_5511  
  top_k: 6  
  score_threshold: 0.72
```

7. Frequently asked questions

What is the practical difference between fixed-size and semantic chunking?

Fixed-size chunkers slice the source text into windows of N tokens or characters, with an overlap between windows to preserve context across boundaries. They are cheap, deterministic, and language-agnostic. Semantic chunkers embed sentences and cluster them by similarity, so adjacent sentences on the same topic land in the same chunk regardless of how many characters they occupy. Semantic chunking usually wins on retrieval quality when the corpus contains long-form documents with multiple topics interleaved. Fixed-size wins on ingest speed and predictability.

Does chunk size affect retrieval quality?

Yes, but the relationship is not monotonic. Very small chunks fragment context and force the retriever to return more chunks per query, which inflates the LLM prompt and the latency budget. Very large chunks dilute the embedding: the vector represents an average of multiple topics, so relevant queries score lower than they should. In practice, chunks between roughly two hundred and eight hundred tokens perform best on general-purpose documentation.

How much overlap should adjacent chunks share?

Ten to twenty percent of chunk size is a defensible starting point. Higher overlap helps recall at the cost of storage and duplication in the retrieved set. If your evaluation set includes questions whose answers span chunk boundaries, err on the higher end.

Should the chunker's tokenizer match the embedder's tokenizer?

Only loosely. The chunker's tokenizer is used to enforce a size budget so chunks fit inside the embedder's input window. It does not need to match byte-for-byte, but it should not under-count either — you do not want the chunker to emit a "five hundred token" chunk that the embedder considers seven hundred tokens and truncates.

8. Case study: a mid-market SaaS support desk

A B2B SaaS company with roughly forty thousand active accounts operated a tier-one support desk staffed by twelve human agents on rotating shifts. Call volume averaged fourteen hundred calls per week with pronounced Monday-morning spikes; average handle time sat at seven minutes forty seconds. Customer satisfaction scores had been trending down for three consecutive quarters, driven almost entirely by wait times during peak hours.

The company deployed a voice agent as a first-line responder in the same phone tree the human agents used, with escalation triggers wired to sentiment detection and to explicit customer requests for a human. The agent was equipped with a retrieval-augmented knowledge base consisting of eleven hundred help articles, a policy handbook, and a rolling window of the last thirty days of resolved tickets. Chunking used a semantic strategy with a five-hundred token target and a similarity threshold of zero point seven eight.

After twelve weeks in production, the voice agent handled forty-two percent of incoming calls to completion without escalation. Average handle time on those calls was two minutes eleven seconds, roughly a third of the human baseline. Wait times during Monday-morning peaks dropped from an average of six minutes forty seconds to under one minute. Customer satisfaction scores on agent-handled calls rebounded to within a point of the human baseline; on escalated calls the scores were actually higher, because callers reached a human faster.

The most interesting finding was not the top-line efficiency win but the shift in what the human agents spent their time on. Freed from the repetitive "reset my password" and "how do I export my data" queries, the humans took on longer, harder cases: onboarding calls with new customers, retention conversations with at-risk accounts, and post-incident debriefs. Handle time on those calls actually went up. Both the agents and the customers preferred it.

9. Glossary

Term	Definition
Chunk	A contiguous slice of source text emitted by a chunker; the unit that is embedded and stored.
Embedding	A dense vector representation of a chunk produced by an embedding model.
Ingestion pipeline	The end-to-end path from raw upload to indexed vector: parse → chunk → embed → store.
Knowledge base	A named collection of ingested documents that a voice agent can retrieve against.
Latency budget	The end-to-end p50 target from caller utterance end to agent audio start; typically 800 ms.
Overlap	Text repeated between adjacent chunks to preserve context across boundaries.
RAG	Retrieval-Augmented Generation; querying a knowledge base at inference time to ground the LLM.
Recall	Proportion of relevant chunks retrieved out of all relevant chunks that exist for a query.
Semantic chunking	Chunking strategy that groups sentences by embedding similarity rather than by fixed size.
Sentence chunker	Chunker that respects sentence boundaries and packs sentences into a token budget.
Threshold	The cosine-similarity cutoff below which a search hit is discarded from the retrieval result.
Top-k	The number of highest-scoring chunks returned by the vector store per query.

End of test corpus. If this document ingests without warnings and each chunker returns a non-empty, sensible chunk list, the pipeline is healthy.